Keyboard / Keystroke Dynamics

Technical Characteristics and Feature Extraction

Introduction

- Behavioral biometric based on typing patterns
- Every individual has unique typing rhythm
- Used for authentication and security
- Works with normal keyboards

What is Keystroke Dynamics

- Keystroke dynamics (also called typing biometrics) is a behavioural biometric technology that analyses the unique pattern of an individual's typing rhythm — including timing, pressure, and sequence — to establish or verify identity.

Types of Keystroke Dynamics

Static Authentication

- Analyses a fixed text entry (e.g., password)
- User types the same password each time
- Profile built during enrolment phase
- Match against stored template on login
- Low overhead — suitable for existing systems

Continuous / Dynamic Authentication

- Monitors typing throughout the session
- No dedicated login challenge required
- Detects session takeover in real-time
- Profile built from free-text typing
- Higher security but more computational cost

Static vs Continuous Authentication

- Static – authentication during login
- Continuous – monitoring during entire session

Improves security against session hijacking

Core Technical characteristics

- Dwell Time(DT)
  - Duration a key is held down. Measured in milliseconds between key-press and key-release events.
- Flight Time (FT)
  - Time between release of one key and press of the next. Captures transition rhythm between key pairs.
- Key Press Latency (KPL)
  - Time between successive key-press events (press-to-press). Includes dwell + flight time.
- N-Graph(NG)
  - Timing across specific 2-key (digraph) or N-key sequences. Captures muscle memory patterns.
- Error Rate (ER)
  - Frequency and timing of backspace/delete keystrokes — unique to an individual's typing accuracy.
- Words per Minute (WPM)
  - Words per minute; a macro-level rhythmic feature. Varies with content, fatigue, and stress.

Dwell Time Feature

- Time between key press and key release
- Indicates how long a key is held

Flight Time Feature

- Time between releasing one key and pressing next key
- Reflects typing rhythm

Latency Feature

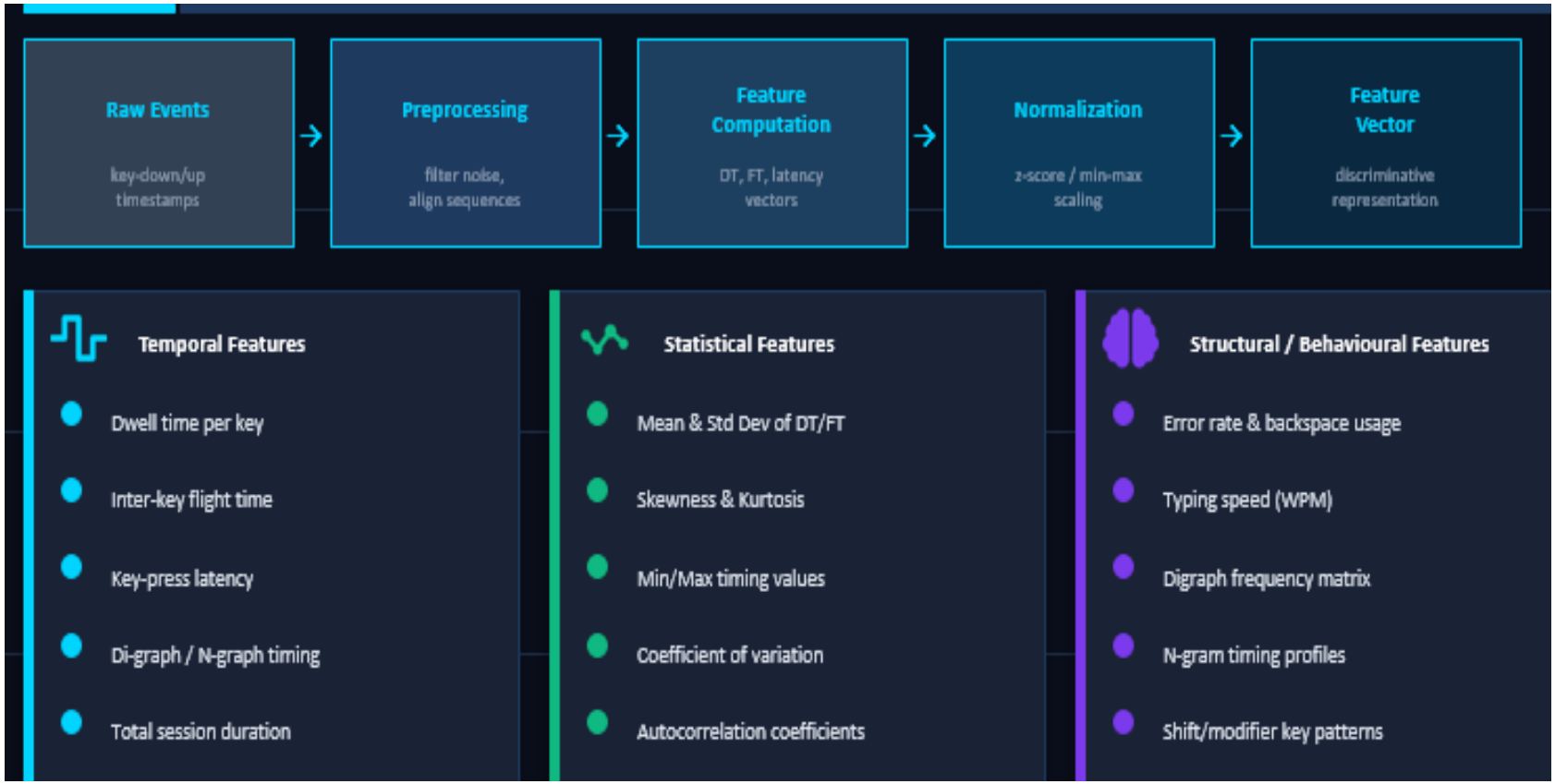
- Delay between successive keystrokes
- Important for rhythm analysis

Data Acquisition Methods

- Hardware-based Capture
  - Keyboard Firmware/controller
  - PS/2 & USB HID Logging
  - Physical Keylogger
  - DevicesTouchscreen sensors
- Software method
  - OS Keyborad Hooks
  - Javascript Event Listeners
  - WSDK/API Libraries
  - Dedicated Biometric SDK’s

Feature Extraction- Overview

- Transforming raw keystroke events into discriminative feature vectors



Temporal Feature Extraction in Detail

Dwell Time (DT_i)

For each key K_i, dwell time is computed as:

$$DT_i = T_release(K_i) - T_press(K_i)$$

Typical range: 40–200 ms per key. Abnormal values are filtered as noise.

Flight Time ($FT_{i,i+1}$)

Between consecutive keys K_i and K_{i+1}:

$$FT_{i,i+1} = T_press(K_{i+1}) - T_release(K_i)$$

Can be negative (overlapping keystrokes during fast typing — important feature itself!).

Digraph & N-graph Timing (Key-Pair Profiles)

Key Pair	Avg DT_1 (ms)	Avg FT (ms)	Std Dev FT
T → H	82	45	12
H → E	88	38	9
E → (space)	75	55	18
(space) → A	91	67	22

Each user develops a unique digraph timing profile reflecting muscle memory. For a password of N characters, N−1 digraph timings are extracted, forming a temporal feature vector. N-graphs extend this to sequences of 3+ characters for richer discrimination.

Unigraph Features

- Timing related to a single key
- Example: dwell time of 'A'

Digraph Features

- Timing relationship between two consecutive keys
- Example: typing 'TH'

Trigraph Features

- Timing across three consecutive keys
- Provides more detailed typing patterns

Statistical Feature Extraction

Mean (μ)

$$\mu = (1/N)\ \Sigma\ x_i$$

Central tendency of dwell/flight times across the sample. Most basic feature — highly interpretable.

Standard Deviation (σ)

$$\sigma = \sqrt{[(1/N)\ \Sigma\ (x_i - \mu)^2]}$$

Spread of timing values. Low σ → consistent typist. High σ → erratic or fatigued typist.

Skewness

$$\gamma_1 = [(1/N)\ \Sigma\ (x_i-\mu)^3]\ /\ \sigma^3$$

Asymmetry of timing distribution. Positive skew → occasional long pauses. Useful for anomaly detection.

Kurtosis

$$\kappa = [(1/N)\ \Sigma\ (x_i-\mu)^4]\ /\ \sigma^4 - 3$$

Peakedness / tail weight of distribution. High kurtosis → most keystrokes very uniform with rare outliers.

Coeff. of Variation

$$CV = (\sigma\ /\ \mu) \times 100\%$$

Normalized dispersion. Allows comparison across users with different baseline speeds.

Autocorrelation

$$r(k) = \Sigma\ (x_i-\mu)(x_{i+k}-\mu)\ /\ \Sigma(x_i-\mu)^2$$

Temporal dependency between successive keystrokes. Captures periodic typing rhythm patterns.

Pre-processing Pipeline Before Feature Extraction

- Event Capture & Timestamping
- Noise filtering
- Sequence Alignment
- Normalization/ Scaling
- Feature Vector Construction

What is Keystroke Dynamics?

- Measures timing information of key events
- Captures press and release actions
- Creates a typing profile for each user

Working Principle

- Capture typing data
- Extract timing features
- Create user template
- Compare during authentication

Keystroke Data Acquisition

- Key press event (Key Down)
- Key release event (Key Up)
- Timestamp recording
- Key sequence tracking

Advanced Feature Extraction Techniques

CNN-based Temporal Pattern Features

- Map keystroke timing sequence as a 1D signal or 2D heatmap image
- Convolutional filters automatically learn discriminative temporal motifs
- No manual feature engineering required
- Captures local & global dependencies simultaneously

LSTM / RNN Sequence Features

- Model the sequential dependencies in keystroke streams
- Hidden state encodes 'typing memory' — past keystrokes influence current
- Effective for continuous (free-text) authentication
- Bidirectional LSTM captures both temporal directions

Transformer / Attention Features

- Self-attention learns which key pairs are most discriminative
- Positional encoding preserves keystroke ordering
- Handles variable-length sequences naturally
- State-of-the-art accuracy on benchmark datasets

PCA, LDA, and Autoencoder-based dimensionality reduction are commonly applied to reduce high-dimensional feature vectors before classification.

Classification Techniques

Statistical Methods

Euclidean Distance	Simple distance from mean profile; baseline method
Mahalanobis Distance	Accounts for covariance between timing features
Gaussian Mixture Model	Models multimodal timing distributions per user

Machine Learning

SVM (Support Vector Machine)	Finds optimal hyperplane in feature space; works well with small samples
k-NN (k-Nearest Neighbours)	Classifies based on proximity to enrolled samples
Random Forest	Ensemble of decision trees; handles noisy timing data

Deep Learning

LSTM Networks	Sequence-aware; models temporal dependencies in keystroke stream
Siamese Networks	One-shot learning; compares two typing samples directly
Transformer Models	Best-in-class on large datasets; attention over key sequences

Anomaly Detection

One-Class SVM	Trained only on genuine user; detects deviations
Isolation Forest	Identifies anomalous typing patterns in free-text streams
Autoencoder	Reconstruction error signals impostors; self-supervised

Performance Evaluation Metrics

FAR

False Acceptance Rate

$$FP / (FP + TN)$$

Probability that an impostor is accepted. Lower = more secure.

FRR

False Rejection Rate

$$FN / (FN + TP)$$

Probability that a genuine user is rejected. Lower = better usability.

EER

Equal Error Rate

$$FAR = FRR \text{ threshold}$$

Operating point where FAR equals FRR. Lower EER = better system.

AUC

Area Under ROC Curve

$$\int_0^1 TPR \, d(FPR)$$

Overall system discriminability. 1.0 = perfect; 0.5 = random.

Published Benchmark EERs

System / Algorithm	Dataset	EER (%)
Mahalanobis Distance (Baseline)	CMU Keystroke Dataset	9.0%
SVM with RBF Kernel	GREYC Dataset	4.5%
LSTM (Deep Learning)	TypeNet Free-text	2.3%
Siamese Network	Aalto Mobile Dataset	1.8%

Advantages

- Difficult to imitate typing rhythm
- Continuous monitoring possible
- Works remotely
- Low implementation cost
- Resilient to stolen passwords-adds ‘something you ARE’

Limitations

- Sensitive to keyboard type changes
- Performance degrade with Fatigue, illness, injury etc.
- Relatively higher EER than physiological biometrics
- Privacy concerns: continuous keystroke monitoring raises GDPR issues.
- Requires training data

Feature Extraction Overview

- Converts raw keystroke data into features
- Captures typing behavior patterns
- Used for classification

Feature Vector Creation

- Combine multiple features
- Create numerical vector
- Used by machine learning models

Feature Normalization

- Remove noise from data
- Normalize typing speed
- Improve system accuracy

Machine Learning Methods

- Support Vector Machines
- Neural Networks
- Random Forest
- k-NN

Applications

- Online Banking & E-Commerce
- Corporate Access Control
- Secure login systems
- Banking fraud detection
- Online exams
- Insider threat detection

Future Trends

- Deep learning for behavior modeling
- Multimodal biometrics
- Mobile typing biometrics